The Three-Bit Plain Heart Inequality

A symmetrization and cycle proof

Abstract

We prove the three-bit plain Heart inequality for uniform one-bit marginals and equal pure-state channels. The proof has four stages:

remove the cubic Fourier coefficient $\longrightarrow$ equalize the two largest masses $\longrightarrow$ equalize the three largest masses $\longrightarrow$ cycles collapse to edges.

The equalization argument uses an exact pair-divided-gradient formula, but no KKT analysis or boundary classification. Its auxiliary estimates reduce to three standard moves: Hölder's inequality, one integration by parts, and a one-derivative Padé bound for artanh. On the final one-small-mass line, the resolvent coefficients are weighted proper colorings of cycles; a uniformizing substitution and Hölder collapse every cycle to its edges. No 2×2 eigenvalue interpolation, crossing-case split, beta-series comparison, or Bernstein-polynomial certificate is needed.

1 Statement and notation

Let u_n denote the uniform law on $\{\pm 1\}^n$. Fix $0 \le s \le 1$, put

$$\mu = \sqrt{1-s^2}, \qquad |\psi_x\rangle = \sqrt{\frac{1+\mu}{2}}\,|0\rangle + x\sqrt{\frac{1-\mu}{2}}\,|1\rangle, \qquad x \in \{\pm 1\},$$

and define the product pure-state channel

$$\Phi_s(p) = \sum_x p(x) \bigotimes_{i=1}^{n} |\psi_{x_i}\rangle\langle\psi_{x_i}|.$$

The uniform input produces

$$\sigma_n = \sigma^{\otimes n}, \qquad \sigma = \operatorname{diag}\left(\frac{1+\mu}{2}, \frac{1-\mu}{2}\right).$$

Independently reveal each coordinate with probability s, and let R be the random revealed set. Define

$$E_s(p) = \mathbb{E}_R D(p_R \| u_R), \qquad Q_s(p) = D(\Phi_s(p) \| \sigma_n), \qquad G_s(p) = E_s(p) - Q_s(p).$$

All logarithms are natural, with $0 \log 0 = 0$.

Theorem 1.1 (Three-bit plain Heart)**.** *If p is a law on $\{\pm 1\}^3$ with uniform one-bit marginals, then*

$$D(\Phi_s(p) \| \sigma_3) \le \mathbb{E}_R D(p_R \| u_R) \qquad (0 \le s \le 1).$$

The endpoints $s = 0, 1$ are equalities. Henceforth assume $0 < s < 1$ and write

$$s = \operatorname{sech}\tau, \qquad \mu = \tanh\tau, \qquad \tau > 0.$$

Uniform one-bit marginals give the Walsh expansion

$$p = u_3\big(1 + a\chi_{12} + b\chi_{13} + c\chi_{23} + d\chi_{123}\big).$$

2 Remove the cubic coefficient

We first show that the three-way Fourier coefficient can only increase the Heart gap.

Lemma 2.1 (Parity-fiber contraction)**.** *Suppose $p - q = k\chi_{123}$ and the segment joining p and q consists of probability laws. Then*

$$D(\Phi_s(p)\|\Phi_s(q)) \le s^3 D(p\|q).$$

Proof. Put

$$r_t = q + tk\chi_{123}, \qquad C_t = \Phi_s(r_t), \qquad H = \frac{\mathrm{d}}{\mathrm{d}t}C_t.$$

For one coordinate,

$$J_i = \frac{|\psi_+\rangle\langle\psi_+| - |\psi_-\rangle\langle\psi_-|}{s}$$

is a Hermitian unitary. Hence, with $J = J_1 \otimes J_2 \otimes J_3$,

$$H = ks^3 J.$$

Let g_C denote the entropy Hessian of $X \mapsto \operatorname{Tr} X \log X$. In an eigenbasis of C_t, the logarithmic-mean bound

$$\frac{\log x - \log y}{x - y} \le \frac{1}{\sqrt{xy}}$$

gives, with $B = C_t^{-1/2} J C_t^{-1/2}$,

$$\begin{aligned} g_{C_t}(H) &\le k^2 s^6 \operatorname{Tr}(BJ) \\ &\le k^2 s^6 \|B\|_1 \|J\| = k^2 s^6 \|B\|_1, \end{aligned}$$

where the second line is trace duality and $\|J\| = 1$.

Write $C_t = \Psi R_t \Psi^*$, with $R_t = \operatorname{diag}(r_t)$ and with the eight product signal vectors as the columns of Ψ. Also

$$J = s^{-3}\Psi E \Psi^*, \qquad E = \operatorname{diag}(\chi_{123}).$$

Now

$$C_t^{-1} J = s^{-3}(\Psi^*)^{-1} R_t^{-1} E \Psi^*,$$

so $C_t^{-1}J$ is similar to $s^{-3}R_t^{-1}E$. Since B is similar to $C_t^{-1}J$ and is Hermitian, its eigenvalues are

$$\frac{\chi_{123}(x)}{s^3 r_t(x)}.$$

Therefore

$$\|B\|_1 = s^{-3} \sum_x \frac{1}{r_t(x)}.$$

Therefore

$$g_{C_t}(H) \le s^3 k^2 \sum_x \frac{1}{r_t(x)} = s^3 \frac{\mathrm{d}^2}{\mathrm{d}t^2} D(r_t \| q).$$

The left side is the second derivative of $D(C_t \| C_0)$. Both divergences and their first derivatives vanish at $t = 0$, so integrating twice proves the claim. Zero atoms follow by approximation. □

Let p_0 be obtained from p by setting $d = 0$. Global complementation changes d to $-d$. Pinching by this symmetry sends $\Phi_s(p)$ to $\Phi_s(p_0)$ and fixes σ_3, hence

$$D(\Phi_s(p) \| \sigma_3) = D(\Phi_s(p) \| \Phi_s(p_0)) + D(\Phi_s(p_0) \| \sigma_3).$$

All proper marginals of p and p_0 agree, so only full reveal sees the cubic coefficient:

$$E_s(p) = E_s(p_0) + s^3 D(p \| p_0).$$

Lemma 2.1 yields

$$G_s(p) \ge G_s(p_0).$$

Thus it is enough to prove the theorem for globally even laws.

3 Even laws are four masses

A globally even law is determined by its four antipodal-pair masses

$$q = (q_1, q_2, q_3, q_4) \in \Delta_4,$$

where

$$q_1 = \frac{1 + a + b + c}{4}, \quad q_2 = \frac{1 + a - b - c}{4}, \quad q_3 = \frac{1 - a + b - c}{4}, \quad q_4 = \frac{1 - a - b + c}{4}.$$

Write $G(q) = G_s(q)$. The classical functional is

$$E_s(q) = s^2(1 - s) \sum_{i<j} (q_i + q_j) \log\big(2(q_i + q_j)\big) + s^3 \sum_i q_i \log(4q_i). \tag{1}$$

The entire minimization will be replaced by two monotone equalizations. The only input is the following pairwise derivative statement.

Lemma 3.1 (The two allowed equalizations)**.** *Let $q_i = x > q_j = y > 0$, and call the other two masses z, u.*

(i) *If $y \ge z, u$, then*

$$\partial_i G > \partial_j G.$$

(ii) *If the masses are (x, x, y, z) with $x > y \ge z$, then, comparing one of the x's with y,*

$$\partial_x G > \partial_y G.$$

We postpone the proof. First we use the lemma to collapse the simplex.

4 Equalization replaces KKT

Relabel so that

$$q_1 \geq q_2 \geq q_3 \geq q_4.$$

First equalize the two largest masses:

$$q(r) = (q_1 - r, q_2 + r, q_3, q_4), \qquad 0 \leq r \leq \frac{q_1 - q_2}{2}.$$

Throughout this path, the selected coordinates remain the two largest. Lemma 3.1(i) gives

$$\frac{\mathrm{d}}{\mathrm{d}r} G(q(r)) = -\partial_1 G + \partial_2 G < 0.$$

Hence

$$G(q_1, q_2, q_3, q_4) \geq G(x, x, q_3, q_4), \qquad x = \frac{q_1 + q_2}{2}. \tag{2}$$

Now equalize the three largest masses:

$$\widetilde{q}(r) = (x - r, x - r, q_3 + 2r, q_4), \qquad 0 \leq r \leq \frac{x - q_3}{3}.$$

This has the form (X, X, Y, q_4) with $X \geq Y \geq q_4$. By symmetry, $\partial_1 G = \partial_2 G$, and Lemma 3.1(ii) gives

$$\frac{\mathrm{d}}{\mathrm{d}r} G(\widetilde{q}(r)) = -\partial_1 G - \partial_2 G + 2\partial_3 G = 2(\partial_3 G - \partial_1 G) < 0.$$

Consequently

$$G(q) \geq G(m, m, m, p), \qquad p = q_4, \quad m = \frac{1 - p}{3}. \tag{3}$$

Thus every law is pushed directly to the one-small-mass line. By symmetry, write that line as

$$q = \left(\frac{1 - 3\rho}{4}, \frac{1 + \rho}{4}, \frac{1 + \rho}{4}, \frac{1 + \rho}{4} \right), \qquad 0 \leq \rho \leq \frac{1}{3}. \tag{4}$$

5 The one-small line: cycles collapse to edges

On the line (4), put

$$h(\rho) = \frac{(1 + \rho)\log(1 + \rho) + (1 - \rho)\log(1 - \rho)}{2}$$

and

$$k(\rho) = \frac{(1 - 3\rho)\log(1 - 3\rho) + 3(1 + \rho)\log(1 + \rho)}{4}.$$

Every two-coordinate reveal has binary bias ρ, while full reveal sees the four masses themselves. Hence

$$E_s(\rho) = 3s^2(1 - s)h(\rho) + s^3 k(\rho). \tag{5}$$

We now bound the quantum term without diagonalizing a single nontrivial block.

5.1 A four-ray entropy contraction

We first isolate the scalar inequality that collapses a cycle edge.

Lemma 5.1 (Edge powers)**.** *For $a, b > 0$ and $p \geq 1$,*

$$\int_0^\infty \left(\frac{a}{a+t}\frac{b}{b+t}\right)^p \,\mathrm{d}t \leq \frac{1}{2p-1}\int_0^\infty \frac{a}{a+t}\frac{b}{b+t}\,\mathrm{d}t. \tag{6}$$

Proof. Write $a = ge^u$ and $b = ge^{-u}$, with $u \geq 0$, and scale $t = gx$. For $0 < v < u$, make the substitution

$$x = \frac{\sinh(u-v)}{\sinh v}.$$

If

$$f(x) = \frac{1}{(x+e^u)(x+e^{-u})},$$

then

$$f(x) = \left(\frac{\sinh v}{\sinh u}\right)^2, \qquad -f(x)\,\mathrm{d}x = \frac{\mathrm{d}v}{\sinh u}.$$

Thus the probability measure proportional to $f(x)\,\mathrm{d}x$ makes v uniform on $[0,u]$, and

$$\frac{\int_0^\infty f(x)^p\,\mathrm{d}x}{\int_0^\infty f(x)\,\mathrm{d}x} = \frac{1}{u}\int_0^u \left(\frac{\sinh v}{\sinh u}\right)^{2p-2}\mathrm{d}v \leq \frac{1}{u}\int_0^u \left(\frac{v}{u}\right)^{2p-2}\mathrm{d}v = \frac{1}{2p-1}.$$

Here we used that $\sinh x/x$ is increasing. The case $u = 0$ follows by continuity. □

For a positive diagonal matrix

$$\Delta = \mathrm{diag}(d_1, d_2, d_3, d_4), \qquad v = (\sqrt{d_1}, \sqrt{d_2}, \sqrt{d_3}, \sqrt{d_4})^T,$$

put

$$K = \Delta - vv^*.$$

Thus K has zero diagonal and $K_{ij} = -\sqrt{d_i d_j}$ for $i \neq j$.

Lemma 5.2 (Cycles-to-edges contraction)**.** *For $0 \leq \rho \leq 1/3$,*

$$D(\Delta + \rho K \| \Delta) \leq \frac{g_\Delta(K)}{3}\,k(\rho), \tag{7}$$

where g_Δ is the entropy Hessian.

Proof. Write

$$F(\rho) = D(\Delta + \rho K \| \Delta).$$

It is enough to treat $0 \leq \rho < 1/3$; the endpoint follows by continuity. For $t \geq 0$, put

$$r_i(t) = \frac{d_i}{d_i + t}$$

and let A_t be the weighted adjacency matrix of the complete graph on four vertices:

$$(A_t)_{ii} = 0, \qquad (A_t)_{ij} = \sqrt{r_i(t) r_j(t)} \quad (i \neq j).$$

Since

$$(\Delta + tI)^{-1/2} K (\Delta + tI)^{-1/2} = -A_t,$$

the resolvent formula for the entropy Hessian gives

$$\begin{aligned} F''(\rho) &= \int_0^\infty \operatorname{Tr}\left[A_t^2 (I - \rho A_t)^{-2}\right] \, \mathrm{d}t \\ &= \sum_{m \geq 0} (m+1)\rho^m \int_0^\infty \operatorname{Tr} A_t^{m+2} \, \mathrm{d}t. \end{aligned} \tag{8}$$

The series converges because every row sum of A_t is at most 3.

We claim that, for every integer $\ell \geq 2$,

$$\int_0^\infty \operatorname{Tr} A_t^\ell \, \mathrm{d}t \leq \frac{N_\ell}{12(\ell-1)} \int_0^\infty \operatorname{Tr} A_t^2 \, \mathrm{d}t, \qquad N_\ell = 3^\ell + 3(-1)^\ell. \tag{9}$$

Indeed, the ℓth resolvent coefficient is a weighted proper-coloring count:

$$\operatorname{Tr} A_t^\ell = \sum_{i_j \neq i_{j+1} \text{ (cyclic)}} \prod_{j=1}^{\ell} r_{i_j}(t).$$

For a fixed proper coloring, set

$$L_{ij} = \int_0^\infty r_i(t) r_j(t) \, \mathrm{d}t.$$

Each vertex occurrence lies on two cycle edges. Hölder's inequality and Lemma 5.1, with $p = \ell/2$, give

$$\begin{aligned} \int_0^\infty \prod_{j=1}^{\ell} r_{i_j}(t) \, \mathrm{d}t &= \int_0^\infty \prod_{j=1}^{\ell} \left(r_{i_j}(t) r_{i_{j+1}}(t)\right)^{1/2} \mathrm{d}t \\ &\leq \frac{1}{\ell - 1} \prod_{j=1}^{\ell} L_{i_j i_{j+1}}^{1/\ell} \\ &\leq \frac{1}{\ell(\ell-1)} \sum_{j=1}^{\ell} L_{i_j i_{j+1}}. \end{aligned}$$

There are N_ℓ proper colorings of C_ℓ with four colors. By color symmetry, each of the 12 directed edges occurs equally often at every fixed position. Summing the last inequality over all colorings proves (9).

Insert (9) into (8). Since

$$\int_0^\infty \operatorname{Tr} A_t^2 \, \mathrm{d}t = g_\Delta(K),$$

we obtain

$$\begin{aligned} F''(\rho) &\leq \frac{g_\Delta(K)}{12} \sum_{m \geq 0} N_{m+2} \rho^m \\ &= \frac{g_\Delta(K)}{(1-3\rho)(1+\rho)} = \frac{g_\Delta(K)}{3} \, k''(\rho). \end{aligned}$$

Both F and k, together with their first derivatives, vanish at 0. Integrating twice proves (7). □

5.2 Apply the contraction to the two parity blocks

Put

$$p_+ = \frac{1+\mu}{2}, \qquad p_- = \frac{1-\mu}{2}.$$

In the even computational basis $(000, 011, 101, 110)$ and the odd basis $(111, 100, 010, 001)$, the output state on (4) is the direct sum

$$C_\rho = C_{\rho,+} \oplus C_{\rho,-},$$

where

$$C_{\rho,\pm} = \Delta_\pm + \rho K_\pm, \qquad K_\pm = \Delta_\pm - v_\pm v_\pm^*,$$

with

$$\Delta_+ = \mathrm{diag}(p_+^3, p_+p_-^2, p_+p_-^2, p_+p_-^2),$$
$$\Delta_- = \mathrm{diag}(p_-^3, p_+^2p_-, p_+^2p_-, p_+^2p_-),$$

and $v_\pm$ the vectors of square roots of the corresponding diagonal entries. At $\rho = 0$ this is exactly $\sigma^{\otimes 3}$. Hence Lemma 5.2 gives

$$Q_s(\rho) \le \frac{Q_s''(0)}{3}\, k(\rho). \tag{10}$$

For $\Delta = \mathrm{diag}(a, b, b, b)$ and $K = \Delta - vv^*$,

$$g_\Delta(K) = 6b + 6ab\frac{\log a - \log b}{a - b}.$$

Using

$$p_+p_- = \frac{s^2}{4}, \qquad p_+ - p_- = \mu, \qquad \frac{p_+}{p_-} = e^{2\tau},$$

we obtain

$$\begin{aligned} Q_s''(0) &= g_{\Delta_+}(K_+) + g_{\Delta_-}(K_-) \\ &= 6(p_+p_-^2 + p_+^2p_-) + 24(p_+^3p_-^2 + p_+^2p_-^3)\frac{\tau}{\mu} \\ &= \frac{3}{2}\left(s^2 + s^4\frac{\tau}{\mu}\right). \end{aligned} \tag{11}$$

Thus

$$Q_s(\rho) \le \frac{s^2}{2}\left(1 + s^2\frac{\tau}{\mu}\right) k(\rho). \tag{12}$$

5.3 Two one-turn scalar comparisons

First, set

$$g(\rho) = 6h(\rho) - k(\rho).$$

A direct differentiation gives

$$g''(\rho) = \frac{3(1-5\rho)}{(1-\rho)(1+\rho)(1-3\rho)}.$$

Thus g'' is positive before $1/5$ and negative after $1/5$. Since

$$g(0) = g'(0) = 0, \qquad g'(\rho) \longrightarrow -\infty \quad (\rho \uparrow 1/3),$$

the function g first increases and then decreases. Its minimum is therefore attained at an endpoint. Finally,

$$g(0) = 0, \qquad g\left(\frac{1}{3}\right) = \log\frac{256}{243} > 0.$$

Hence

$$h(\rho) \geq \frac{1}{6}k(\rho). \tag{13}$$

Second,

$$s\frac{\tau}{\mu} \leq 1,$$

because $s = \operatorname{sech}\tau$, $\mu = \tanh\tau$, and the inequality is exactly $\tau \leq \sinh\tau$. Consequently (12) becomes

$$Q_s(\rho) \leq \frac{s^2(1+s)}{2}k(\rho). \tag{14}$$

On the other hand, (5) and (13) give

$$E_s(\rho) \geq \left(\frac{s^2(1-s)}{2} + s^3\right)k(\rho) = \frac{s^2(1+s)}{2}k(\rho).$$

Together with (14), this proves

$$Q_s(\rho) \leq E_s(\rho)$$

on the one-small-mass line. The two equalizations prove the result for every even law, and Lemma 2.1 restores the cubic coefficient.

The proof is therefore linear: parity contraction, two monotone equalizations, and a cycles-to-edges contraction on the final line.

A The equalization lemma: reserve plus a balanced pair

We prove Lemma 3.1. Choose two distinct positive masses x and y, and call the other two masses z and u. For $c \geq 0$, define

$$L_c(x,y) = \frac{\log(x+c) - \log(y+c)}{x-y}, \qquad C_{x,y;z,u} = L_z(x,y) + L_u(x,y) - L_0(x,y).$$

For $r > 0$, put

$$\mathfrak{d}(r) = \frac{1}{(r+x)(r+y)}, \qquad P(r) = \frac{r(2r+x+y)}{2(r+x)(r+y)},$$
$$\theta(r) = 1 - \frac{r}{2}\left(\frac{1}{r+x} + \frac{1}{r+y} + \frac{1}{r+z} + \frac{1}{r+u}\right).$$

Use the natural channel coordinates

$$\omega = \tanh(2\tau), \qquad p_\pm = \frac{1\pm\mu}{2}, \qquad \kappa = \frac{\mu}{\omega} = \frac{1+\mu^2}{2} = 1 - \frac{s^2}{2},$$
$$A = \frac{1-s}{\omega^2} = \frac{(2-s^2)^2}{4(1+s)}.$$

A.1 The exact divided gradient

Let

$$\Delta = \operatorname{diag}(q_1, q_2, q_3, q_4), \qquad v = (\sqrt{q_1}, \sqrt{q_2}, \sqrt{q_3}, \sqrt{q_4})^T,$$

and

$$\eta_\varepsilon = \frac{e^{4\varepsilon\tau} - 1}{4}, \qquad N_\varepsilon = \Delta + \eta_\varepsilon v v^*.$$

The Hadamard decomposition of the two parity blocks shows that, up to a term depending only on $\sum_i q_i = 1$, the quantum functional is

$$Q_s(q) = \frac{s^3}{2} \sum_{\varepsilon=\pm1} e^{-\varepsilon\tau} \operatorname{Tr} N_\varepsilon \log N_\varepsilon. \tag{15}$$

For completeness, this follows from $U^* M U = 4\Delta$ and $U^* H_0 U = 4pp^* - I$, where U is the normalized Hadamard matrix, $H_0 = \operatorname{diag}(3, -1, -1, -1)$, and $p = (1, 1, 1, 1)^T/2$. The nonzero spectra of the two parity blocks and $4e^{-\varepsilon\tau} N_\varepsilon$ coincide; all discarded terms are affine and symmetric in q, hence depend only on $\sum_i q_i$.

The classical formula (1) gives

$$\frac{\partial_i E_s - \partial_j E_s}{x - y} = s^2(1 - s)(L_z + L_u) + s^3 L_0. \tag{16}$$

Moreover,

$$\det(N_\varepsilon + rI) = \prod_k (r + q_k) \left(1 + \eta_\varepsilon \sum_k \frac{q_k}{r + q_k}\right), \qquad \sum_k \frac{q_k}{r + q_k} = 2(1 + \theta(r)).$$

Differentiation in the trace-zero direction $e_i - e_j$ yields

$$-\frac{(\partial_i - \partial_j) \log\det(N_\varepsilon + rI)}{x - y} = \mathfrak{d}(r) \left[1 + \frac{2\eta_\varepsilon P(r)}{1 + 2\eta_\varepsilon(1 + \theta(r))}\right].$$

Using the equal-trace identity

$$\operatorname{Tr} X \log X - \operatorname{Tr} Y \log Y = -\int_0^\infty \log \frac{\det(X + rI)}{\det(Y + rI)}\, \mathrm{d}r$$

and the elementary simplification

$$\sum_{\varepsilon=\pm1} \frac{e^{-\varepsilon\tau}\eta_\varepsilon}{1 + 2\eta_\varepsilon(1 + \theta)} = -\frac{\omega^2}{s} \frac{\theta + \kappa}{1 - \omega^2\theta^2},$$

we obtain

$$\boxed{\frac{\partial_i G - \partial_j G}{x - y} = s^2\omega^2 \left[AC_{x,y;z,u} + \int_0^\infty \mathfrak{d}(r) P(r) \frac{\theta(r) + \kappa}{1 - \omega^2\theta(r)^2}\, \mathrm{d}r\right].} \tag{17}$$

A.2 Split off the complementary reserve

Put

$$F_s(t) = \frac{t+\kappa}{1-\omega^2 t^2}, \qquad S(r) = \frac{r}{2}\left(\frac{1}{r+z} + \frac{1}{r+u}\right).$$

Then

$$\theta = -P + (1-S), \qquad 1 - S = \frac{1}{2}\left(\frac{z}{r+z} + \frac{u}{r+u}\right) \geq 0.$$

Let

$$\mathrm{d}\nu(r) = \mathfrak{d}(r)P(r)\ \mathrm{d}r, \qquad V = \int_0^\infty \mathrm{d}\nu = \frac{1}{2}L_0, \qquad J = \int_0^\infty (1-S)\ \mathrm{d}\nu.$$

The identity for V follows from $P = -r\mathfrak{d}'/(2\mathfrak{d})$ and integration by parts.

The channel kernel has the partial-fraction form

$$F_s(t) = \frac{1}{\omega}\left(\frac{p_+}{1-\omega t} - \frac{p_-}{1+\omega t}\right), \tag{18}$$

so

$$F_s'(t) = \frac{p_+}{(1-\omega t)^2} + \frac{p_-}{(1+\omega t)^2}. \tag{19}$$

For $a, b, X, Y > 0$, Hölder's inequality gives

$$\frac{a}{X^2} + \frac{b}{Y^2} \geq \frac{(a^{1/3}+b^{1/3})^3}{(X+Y)^2}.$$

Apply this with $a = p_+$, $b = p_-$, $X = 1-\omega t$, and $Y = 1+\omega t$. Since $X + Y = 2$,

$$F_s'(t) \geq \lambda_s := \frac{1}{4}\big(p_+^{1/3} + p_-^{1/3}\big)^3. \tag{20}$$

Therefore

$$\int F_s(\theta)\ \mathrm{d}\nu \geq \int F_s(-P)\ \mathrm{d}\nu + \lambda_s J. \tag{21}$$

A.3 The reserve inequality

Lemma A.1. *For all $x, y > 0$ and $z, u \geq 0$,*

$$C_{x,y;z,u} + 6J \geq L_0 = 2V. \tag{22}$$

Proof. For $c \geq 0$, set

$$J_c = \int_0^\infty \mathfrak{d}(r)P(r)\frac{c}{r+c}\ \mathrm{d}r.$$

Then $J = (J_z + J_u)/2$. Since $P = -r\mathfrak{d}'/(2\mathfrak{d})$, integration by parts gives

$$J_c = \frac{c^2}{2}\int_0^\infty \frac{\mathfrak{d}(r)}{(r+c)^2}\ \mathrm{d}r. \tag{23}$$

Also

$$L_0 - L_c = \int_0^c \mathfrak{d}(r)\ \mathrm{d}r.$$

We claim

$$L_0 - L_c \leq 3J_c. \tag{24}$$

The case $c = 0$ is trivial. For $c > 0$, use

$$\mathfrak{d}(r) = \int_0^1 \frac{\mathrm{d}\alpha}{(r + (1-\alpha)x + \alpha y)^2}.$$

It is therefore enough to prove, for $a, c > 0$,

$$\int_0^c \frac{\mathrm{d}r}{(r+a)^2} \le \frac{3c^2}{2} \int_0^\infty \frac{\mathrm{d}r}{(r+a)^2(r+c)^2}. \tag{25}$$

Scale $c = 1$ and put $q = (a-1)/(a+1) \in (-1, 1)$. Direct integration turns the difference between the right and left sides into the positive prefactor

$$\frac{(1-q)^2}{4(1+q)}$$

times

$$\frac{q(3-2q^2) - 3(1-q^2)\operatorname{artanh} q}{q^3}.$$

Define

$$\phi(q) = \frac{q(3-2q^2)}{3(1-q^2)} - \operatorname{artanh} q.$$

Then

$$\phi(0) = 0, \qquad \phi'(q) = \frac{2q^4}{3(1-q^2)^2} \ge 0.$$

Hence $\phi(q)/q \ge 0$ for $q \ne 0$, and the last displayed quotient is nonnegative. This proves (25), hence (24). Summing for $c = z, u$ gives

$$(L_0 - L_z) + (L_0 - L_u) \le 3(J_z + J_u) = 6J,$$

which is exactly (22). □

A.4 The balanced-pair comparison

Lemma A.2. *Define*

$$\mathcal{B}_s = 2 \int_0^1 p F_s(-p)\, \mathrm{d}p. \tag{26}$$

Then

$$\int_0^\infty F_s(-P(r))\, \mathrm{d}\nu(r) \ge V \min\{\mathcal{B}_s, 0\}. \tag{27}$$

Proof. Normalize ν to a probability measure and write $m_k = \mathbb{E}_\nu P^k$. Put

$$\gamma = \left(\frac{x-y}{x+y}\right)^2, \qquad g_\gamma(v) = 1 - \gamma(1-v)^2.$$

With $d = (x+y)/2$ and $r = dv/(1-v)$,

$$\int_0^\infty \mathfrak{d}(r) P(r)^{k+1}\, \mathrm{d}r = \frac{1}{d} I_k(\gamma), \qquad I_k(\gamma) = \int_0^1 \frac{v^{k+1}}{g_\gamma(v)^{k+2}}\, \mathrm{d}v.$$

Instead of expanding in powers of γ, integrate the derivative of

$$\frac{v^{k+2}(1-v)}{g_\gamma(v)^{k+2}}.$$

The boundary terms vanish, and direct simplification gives

$$(k+2)I_k - (k+3)I_{k+1} = \gamma(k+1)\int_0^1 \frac{v^{k+2}(1-v)^2}{g_\gamma(v)^{k+3}}\,\mathrm{d}v \geq 0. \tag{28}$$

Consequently

$$\frac{m_{k+1}}{m_k} = \frac{I_{k+1}}{I_k} \leq \frac{k+2}{k+3}. \tag{29}$$

Applying this for $k = 2n$ and $k = 2n+1$ shows that

$$d_n = (n+1)m_{2n}$$

is decreasing, with $d_0 = 1$.

Now expand

$$F_s(-P) = \sum_{n\geq 0} \omega^{2n}\big(\kappa P^{2n} - P^{2n+1}\big).$$

By (29), with $r_n = (2n+2)/(2n+3)$,

$$\mathbb{E}_\nu F_s(-P) \geq \sum_{n\geq 0} a_n d_n, \qquad a_n = \frac{\omega^{2n}}{n+1}(\kappa - r_n).$$

The sequence r_n increases, so the a_n change sign at most once, from positive to negative. The one-signed cases are immediate. Otherwise, if N is the crossing index, monotonicity of d_n gives

$$\sum_n a_n d_n \geq d_N \sum_n a_n.$$

If the last sum is nonnegative, this is at least 0; if it is negative, $d_N \leq 1$ makes it at least $\sum_n a_n$. Hence

$$\sum_n a_n d_n \geq \min\left\{\sum_n a_n, 0\right\}.$$

For $x = y$, one has $m_k = 2/(k+2)$ and therefore $d_n = 1$. Thus

$$\sum_n a_n = 2\int_0^1 pF_s(-p)\,\mathrm{d}p = \mathcal{B}_s.$$

Multiplying by V proves (27). □

A.5 The remaining channel scalar

Lemma A.3. *For $0 < s < 1$,*

$$\min\{\mathcal{B}_s, 0\} + 2\min\left\{A, \frac{\lambda_s}{6}\right\} > 0. \tag{30}$$

Proof. We use four coarse estimates.

First,

$$A = \frac{(2-s^2)^2}{4(1+s)} \geq \frac{1}{8}. \tag{31}$$

Indeed, if

$$f(s) = 2(2-s^2)^2 - (1+s),$$

then

$$f'(s) = -8s(2-s^2) - 1 < 0, \qquad f(1) = 0.$$

Second, writing $p = p_+^{1/3}$ and $q = p_-^{1/3}$,

$$4\lambda_s = (p+q)^3 = 1 + 3pq(p+q) \geq 1 + 6(pq)^{3/2} = 1 + 3s.$$

Thus

$$\lambda_s \geq \frac{1+3s}{4}. \tag{32}$$

Third,

$$\mathcal{B}_s \geq -\frac{1}{5}. \tag{33}$$

To see this, write

$$\omega = \frac{2\mu}{1+\mu^2}, \qquad L = \log\frac{1+\mu^2}{1-\mu^2}.$$

Direct integration in (26) gives

$$\mathcal{B}_s = \frac{2}{\omega^3}\mathcal{N}(\mu), \qquad \mathcal{N}(\mu) = \omega - 2\operatorname{artanh}\mu + \mu L. \tag{34}$$

The derivative collapses to

$$\mathcal{N}'(\mu) = L - \omega^2. \tag{35}$$

Set

$$\mathcal{M}(\mu) = \mathcal{N}(\mu) + \frac{\omega^3}{10}, \qquad x = \mu^2.$$

Then

$$\mathcal{M}'(\mu) = L - \frac{4x}{(1+x)^2} + \frac{12x(1-x)}{5(1+x)^4}.$$

Since

$$L = 2\operatorname{artanh} x \geq 2x,$$

we get

$$\mathcal{M}'(\mu) \geq \frac{2x}{5(1+x)^4}\left(5x^4 + 20x^3 + 20x^2 - 6x + 1\right) > 0,$$

because

$$5x^4 + 20x^3 + 20x^2 - 6x + 1 \geq 20x^2 - 6x + 1 > 0.$$

Since $\mathcal{M}(0) = 0$, this proves $\mathcal{N} \geq -\omega^3/10$, which is (33).

Fourth,

$$s \leq \frac{1}{2} \quad \Longrightarrow \quad \mathcal{B}_s > 0. \tag{36}$$

Indeed,

$$2\operatorname{artanh}\mu = \operatorname{artanh}\omega = L + \log(1+\omega),$$

so

$$\mathcal{N} = \omega - \log(1+\omega) - (1-\mu)L. \tag{37}$$

If $s \le 1/2$, then

$$\mu \ge \frac{\sqrt{3}}{2} > \frac{6}{7}.$$

Put $y = 1 - \mu < 1/7$. Since

$$\frac{1+\mu^2}{1-\mu^2} \le \frac{1}{y},$$

we have

$$(1-\mu)L \le -y \log y \le \frac{\log 7}{7} < \frac{2}{7}.$$

Also $\omega > 84/85$, and $r \mapsto r - \log(1+r)$ is increasing. Therefore

$$\omega - \log(1+\omega) > \frac{84}{85} - \log 2 > \frac{84}{85} - \frac{7}{10} = \frac{49}{170} > \frac{2}{7}.$$

Equation (37) gives $\mathcal{N} > 0$, proving (36).

If $\mathcal{B}_s \ge 0$, then (30) is immediate. Otherwise (36) implies $s > 1/2$. Equations (31) and (32) then give

$$2A \ge \frac{1}{4} > \frac{1}{5}, \qquad \frac{\lambda_s}{3} > \frac{5}{24} > \frac{1}{5}.$$

Together with (33), this proves (30). □

A.6 Finish the ordered-pair criterion

Assume $C_{x,y;z,u} \ge 0$. Combining (21), Lemma A.1, and Lemma A.2, the bracket in (17) is at least

$$\begin{aligned} AC_{x,y;z,u} + \lambda_s J + V \min\{\mathcal{B}_s, 0\} &\ge 2V \min\left\{A, \frac{\lambda_s}{6}\right\} + V \min\{\mathcal{B}_s, 0\} \\ &> 0 \end{aligned}$$

by Lemma A.3. Therefore

$$C_{x,y;z,u} \ge 0 \quad \Longrightarrow \quad \frac{\partial_i G - \partial_j G}{x - y} > 0. \tag{38}$$

It remains only to verify the reserve in the two configurations used by the equalization paths. If $x > y \ge z, u$, then $c \mapsto L_c(x,y)$ decreases, and

$$C_{x,y;z,u} \ge 2L_y - L_0 = \frac{1}{x-y} \log \frac{(x+y)^2}{4xy} > 0.$$

If the masses are (x, x, y, z) with $x > y \ge z$, and the selected pair is one x and the mass y, then

$$C_{x,y;x,z} = \frac{1}{x-y} \log \frac{2y(x+z)}{(x+y)(y+z)}.$$

The numerator minus the denominator inside the logarithm is

$$2y(x+z) - (x+y)(y+z) = (x-y)(y-z) \ge 0.$$

Thus the reserve is nonnegative in both cases, proving Lemma 3.1. Vanishing masses follow by approximation.